

HANNAH GUAN

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EDUCATION

Harvard University, B.A. in Applied Math in CS and Economics Cambridge, MA | 2023-2027

GPA 3.867/4.0. *Head Teaching Fellow*: Machine Learning. *Teaching Fellow*: Unsupervised Learning, Algorithms and Limitations. *Graduate Level*: AI Alignment and Safety, Interface between Computer Science and Economics, Reinforcement Learning, Topics in Machine Learning: Learning: Effective AI Support in Human+AI Settings

Basis San Antonio Shavano, Valedictorian San Antonio, TX | 2019-2023

GPA 4.91/5.0, 4.0/4.0; 14 APs all 5s; ACT 36/36 (CEO commendation letter for ACT perfect score).

Trinity University, Coursework taken in high school San Antonio, TX | 2021-2022

GPA 4.0/4.0 (earned 21 college credits in computer science through a Trinity University scholarship).

RESEARCH INTERESTS

AI and Machine Learning Optimization for predictive modeling, decision-making, and pattern recognition across various domains, Computational Mathematics (numerical + optimization), Advanced Statistical Modeling, Causal Inference, Time Series Analysis, Stochastic Differential Equations and Applications, Randomization Methods.

RESEARCH PUBLICATIONS

- **Guan H**, Mouatadid S, Orenstein P, Cohen J, Dong H, Flaspohler G, Lu A, Weyn J, and Mackey L, Advancing Subseasonal Forecasting with Machine Learning, *Nature*, 2026 (forthcoming).
- **Guan H**, Dhanani A, Zhang Y, HiDiNet: High-Dimensional Interpretive Network for Modeling Aging Health and Survival, *ACM Digital Government: Research and Practice*, 7(3), 2026. DOI: <https://doi.org/10.1145/3833874>. Selected for ACM Showcase on Kudos: <https://link.growkudos.com/1ed7qtze3gg>.
- **Guan H**, Mouatadid S, Orenstein P, Cohen J, Dong H, Flaspohler G, Lu A, Weyn J, and Mackey L, Bridging the “Predictability Desert”: A Probabilistic Bias Correction Framework for AI and Dynamical Subseasonal Forecasts, *29th Annual Conference on Artificial Intelligence and Statistics (AISTATS) Workshop on Calibration for Modern AI*, Tangier, Morocco, May 2-5, 2026.
- **Guan H**, Mouatadid S, Orenstein P, Cohen J, Dong H, Flaspohler G, Lu A, Weyn J, and Mackey L, Bridging the “Predictability Desert”: A Probabilistic Bias Correction Framework for AI and Dynamical Subseasonal Forecasts, *14th International Conference on Learning Representations (ICLR) Workshop on Machine Learning for Remote Sensing*. Selected for Oral Spotlight. Rio de Janeiro, Brazil, April 26-27, 2026.
- Cakar E, **Guan H**, and Kehe K, Accelerating Adversarial Suffix Optimization via Continuous Relaxation and Activation-Guided Objectives, *14th International Conference on Learning Representations (ICLR) Workshop on Representational Alignment*. Selected for Long Paper. Rio de Janeiro, Brazil, April 26-27, 2026.
- **Guan H**, Mouatadid S, Orenstein P, Cohen J, Dong H, Flaspohler G, Lu A, Weyn J, and Mackey L, Bridging the “Predictability Desert”: A Probabilistic Bias Correction Framework for AI and Dynamical

Subseasonal Forecasts, *European Geosciences Union General Assembly (EGU)*. Selected for Oral Spotlight. Vienna, Austria, May 3-8, 2026.

- **Guan H** and Hou C, When Debate Fails: An Empirical Study of Incentive Misalignment in Prover-Estimator Games, *Proceedings of the AAAI Symposium Series*, 9(1): 43-51, 2026. DOI: <https://doi.org/10.1609/aaais.v9i1.42904>.
- Cakar E, **Guan H**, and Kehe K, Optimizing Against Safety Representations: Activation-Guided Adversarial Suffixes and the Geometry of Refusal, *Proceedings of the AAAI Symposium Series*, 9(1), 27-34, 2026. DOI: <https://doi.org/10.1609/aaais.v9i1.42902>.
- **Guan H**, Mouatadid S, Orenstein P, Cohen J, Dong H, Flaspohler G, Lu A, Weyn J, and Mackey L, Bridging the “Predictability Desert”: A Probabilistic Bias Correction Framework for AI and Dynamical Subseasonal Forecasts, *Third International Conference on Subseasonal to Seasonal to Decadal Prediction (S2S2D)*, Reading, UK, September 7-11, 2026.
- **Guan H**, Yang A, Yun H, Wang X, Loffree X, Li J, Liang L, Exploring AI Methods for Developing Metabolomic Signatures of Nutritional Factors, *American Society of Human Genetics Annual Meeting (ASHG)*, October 14-18, 2025.
- **Guan H**, Dhanani A, Interpretable Deep Learning for Aging Dynamics and Mortality Prediction, *American Society of Human Genetics Annual Meeting (ASHG)*, October 14-18, 2025.
- **Guan H**, Greiner G, and Martinez L, Innovative Method for Longitudinal Analysis of Aging Health and Mortality Rate Prediction. *Proceedings of the Joint Statistical Meetings (JSM)*, 2024. DOI: doi.org/10.5281/zenodo.14004714.
- **Guan H**, A Novel Approach for Longitudinal Modeling of Aging Health and Predicting Mortality Rates, *Proceedings of the AAAI Conference on Artificial Intelligence Undergraduate Consortium*, 38(21): 23744-23746, 2024. DOI: doi.org/10.1609/aaai.v38i21.30550.
- **Guan H** and Zhang C, Predicting Diabetes in Imbalanced Datasets using Neural Networks, *13th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB)*, 57: 1-10, 2022. DOI: doi.org/10.1145/3535508.3545540.
- Li L, Zhang C, Liu S, **Guan H**, Zhang Y, Age Prediction by DNA Methylation in Neural Networks, *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 19(3): 1393-1402, 2022. DOI: [10.1109/TCBB.2021.3084596](https://doi.org/10.1109/TCBB.2021.3084596).
- Li L, Zhang C, **Guan H**, Zhang Y, Application of Correlation Pre-Filtering Neural Network to DNA Methylation Data: Biological Aging Prediction, *Epigenome-Wide Association Studies: Methods and Protocols*, **Springer Nature**, 2432: 201-210, 2022. DOI: doi.org/10.1007/978-1-0716-1994-0.
- **Guan H**, The Genetics of Human Aging: Predicting Age and Age-Related Diseases by Deep Mining High Dimensional Biomarker Data, *12th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB)*, 79, 2021. DOI: doi.org/10.1145/3459930.3469508.
- **Guan H**, Genetic Prediction of Biological Age: Exploring the Relationship between Epigenetic Markers and All-Cause Mortality, *Biomedical Science*, 34(2): 26546-26552, 2021. DOI: [BJSTR.MS.ID.005518](https://doi.org/10.1007/978-1-0716-1994-0).
- **Guan H**, Effect of Fish Oil on Heart Health: A Meta-Analysis, *Biomedical Science*, 34(3): 26701-26705, 2021. DOI: [BJSTR.MS.ID.005543](https://doi.org/10.1007/978-1-0716-1994-0).

RESEARCH AWARDS

- 2025 1st-Place Global Winner of the European Centre for Medium-Range Weather Forecasts (ECMWF) AI Weather Quest. MicroEnsemble placed first for each target weather variable: temperature (tas), mean sea-level pressure (mslp), and precipitation (pr), and each forecast horizon, outperforming the operational dynamical models of six government forecasting agencies, the in-house AI models of ECMWF, and the forecasting systems of 166 participants from 14 countries.
- 2025 CRA (Computing Research Association) Outstanding Undergraduate Researcher Awards Winner (Top Computing honor). First sophomore winner in 24 years since the program’s inception.
- 2025 ACM (Association of Computing Machinery) Student Research Competition Grand Finalist (1 of 15 globally).

- 2024 NCWIT (National Center for Women and Information Technology) Collegiate Award Winner (National Top 1 female college researcher in Computing).
- 2024 Heidelberg Laureate Forum Fellow (World Top 200 young researcher in Math and Computing).
- 2024 First-Place Winner in the ACM Student Research Competition in the 2024 ACM Tapia Conference.
- 2024 AAAI Undergraduate Consortium Scholar: Awarded a full scholarship for presenting paper in the 38th AAAI Conference on AI. First freshman scholar since the program's inception.
- 2023 82nd Regeneron Science Talent Search Top 40: Nation's oldest and most prestigious pre-collegiate science competition. The first finalist in 33 years in San Antonio, TX, the 7th largest city in the US.
- 2023 ACM Cutler-Bell Prize in High School Computing Winner: World's largest computing award to high school students. 1 of 4 global winners.
- 2022 International Science and Engineering Fair (ISEF) 4th Place Grand Prize Winner (World Top 20) & 2nd Place American Statistical Association Special Award Winner.
- 2022 Atlas Fellow: A global talent search for original thinkers of all backgrounds. Selected for dedication to solving the world's most pressing problems in health and medicine.
- 2021 ACM BCB Best Poster Award at the ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics, the flagship conference of the ACM SIGBio.
- 2021 MathFest Outstanding Undergraduate Presentation Award by the Mathematics Association of America.
- 2021 Spirit of Ramanujan Fellow: A global STEM talent search recognized alongside the MacArthur Fellowship.
- 2021 World Science Scholar: A two-year learning and discovery program alongside Nobel and Breakthrough Prize winners. 48 scholars from 25 countries.
- 2021 MAA Math Values Featured Researcher on September 30, 2021.
- 2021 ACM Digital Library Featured Researcher on August 27, 2021.

ACADEMIC AWARDS

- 2025 2nd-Place Winner, Citadel Securities Trading Invitational (Top 2 of 90 finalists selected from 3,000+ applicants representing top universities worldwide).
- 2024 Harvard First-Year Veritas Award Finalist (One of three finalists out of 1,942 freshmen).
- 2024 3rd-Place Winner, Citadel Women's Datathon (Top 3 of 80 finalists selected from thousands of applicants from top universities worldwide).
- 2023 National Top 5 Young Women in Mathematics Winner by Mathematical Association of America (MAA).
- 2021-2023 3X USA Mathematical Olympiad (National Top 0.07%).
- 2022-2023 2X USA Physics Olympiad (2X, National Top 0.3%).
- 2022 Regeneron International Science and Engineering Fair (ISEF) 4th Place Grand Prize Winner & 2nd Place American Statistical Association Special Award Winner (World Top 20), SR. Joseph Marie Armer Award for Distinguished Leadership in Science.
- 2021 USA Computing Olympiad Gold Division (National Top 0.5%).
- 2021 National Center for Women & Information Technology Aspirations in Computing Winner (National Top 40).
- 2021 USA Math Talent Search Bronze Prize (National Top 30).
- 2021 Top 1 girl & Top 4 overall in American Mathematics Competitions (330,000 entrants).
- 2021 Best Poster Award at the 12th ACM International Conference on Bioinformatics, and Computational Biology (350 researchers from 26 countries).
- 2021 Outstanding Undergraduate Presentation Award at the MAA MathFest Conference (986 undergraduates from 309 universities).
- 2021 Diagnostics Research Group Special Award.
- 2021 New York Academy of Sciences Junior Academy Young Member.
- 2021 American Society of Human Genetics Young Member.

- 2020 1st Place MIT COVID International Hackathon (5000 global entrants).
- 2019 Minor Planet 12885 Namesake by MIT Lincoln Lab.
- 2019 Lemelson Early Inventor Prize (Top 1 project in Texas State Science and Engineering Fair).
- 2019 Naval Science Award (Top 1 project in Alamo Regional Science and Engineering Fair).
- 2018 U.S. Team and Co-Champions at Primary Mathematics World Competition (World Top 4).
- 2017 Duke TIP (Duke University Talent Identification Program) Texas Grand Award Winner (Highest SAT Score in Texas (7th Grade)).

LEADERSHIP AWARDS

- 2026 United Nations Millennium Fellowship (One of 5,500 fellows selected with 76,000+ applicants from 6,000+ campuses over 170 nations).
- 2026 Harvard University Women's Leadership Impact Award.
- 2025 U.S. Presidential Volunteer Service Lifetime Achievement Award.
- 2024 Carson Scholar Hall of Fame Inductee (Top Honor in Carson Scholar Foundation).
- 2024 AnitaB.org Advancing Inclusion Scholar.
- 2023 U.S. Presidential Scholar. 161 scholars, 3.7M entrants.
- 2023 Coolidge Presidential Scholar: A Full Ride scholarship applies to any university with any major. 5 scholars, 12,000 entrants.
- 2023 Coca-Cola Scholar: Nation's largest corporate-sponsored scholarship. 150 scholars, 91,000 entrants.
- 2023 Grace Hopper Celebration Scholar.
- 2022 Cameron Impact Scholar: A Full Tuition scholarship applies to any university with any major. 15 scholars, 3,500 entrants. Declined in favor of the Coolidge Presidential Scholarship.
- 2022 UN Global Student Prize Top 50 (Selected from 3500 nominees in 94 countries, judged on a rigorous set of criteria to identify extraordinary students).
- 2022 Gloria Barron Prize for Young Heroes Winner (World Top 15).
- 2022 Princeton Prize in Race Relations Winner (World Top 29).
- 2022 Yale Bassett Award for Community Engagement Winner (World Top 20).
- 2021-2022 2X Carson Scholar, featured in the Carson Foundation Original Movie – Heroes Among Us.
- 2021 National Liberty Museum Young Heroes Winner (National Top 15).
- 2021 President George H. W. Bush Daily Point of Light Honoree.
- 2021 Prudential Spirit of Community State Honoree (Top 1 youth volunteer in TX).
- 2021 Governor's Volunteer Award (Top 1 youth volunteer in TX).
- 2021-2022 2X Power of Youth Challenge Winner.
- 2021 Global Youth Service Day Award.
- 2020 Business Journal Nonprofit & Corporate Philanthropy Award.
- 2022 San Antonio's Extraordinary Women.
- 2020 Mayor's Compassionate San Antonian Award.
- 2019-2025 6X Presidential Volunteer Service Award Gold Medal.

WORK EXPERIENCE

Harvard Kempner Institute of Natural and Artificial Intelligence

Research Fellow

Cambridge, MA
September 2025-Present

- Pioneered a modular alignment framework extending Multi-Agent Verification with ranked preferences and social-choice aggregation to produce scalable, trustworthy LLM outputs without retraining.
- Advanced research on trustworthy AI for high-stakes applications in healthcare, climate modeling, and policy through principled decision aggregation.

Microsoft Research

Cambridge, MA

- Research Intern* June 2025-Present
- Built the first global probabilistic subseasonal forecasting model for the ECMWF (European Center for Medium-Range Weather Forecasts) AI Weather Quest, an international competition designed to harness AI and Machine Learning in advancing weather forecasting.
 - Collaborated with Microsoft Research, Microsoft Weather, MIT, U. of Toronto, and Instituto de Matemática Pura e Aplicada (in Brazil) on climate forecasting systems.
- Susquehanna International Group (SIG)** Bala Cynwyd, PA
Quantitative Trading Intern June - August 2026
- Developed and backtested quantitative trading models using high-dimensional market data to optimize options pricing strategies and market-making execution.
 - Leveraged game theory and statistical modeling to analyze complex derivatives data, boosting prediction accuracy for intraday volatility..
- Harvard T. H. Chan School of Public Health** Cambridge, MA
Research Assistant August 2023 - May 2025
- Statistical modeling and machine learning in Python and R, training 2B genetic data in predicting age-related diseases.
 - Developed and fine-tuned various neural network architectures, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Transformers, and Variational Autoencoders (VAEs). Utilized grid search for hyperparameter optimization and applied L1 and L2 regularization techniques. Model validation was conducted using k-fold cross-validation, confusion matrix analysis, ROC Curve and Area Under the Curve, Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) metrics.
- CS 1810 Intro to Machine Learning** Cambridge, MA
Head Teaching Fellow August 2025-Present
- CS 120 Intro to Algorithms and Their Limitations**
STAT 185 Unsupervised Learning
- Teaching Fellow*
- Facilitated student learning through leading discussion sections, hosting office hours, grading assignments, and providing feedback on core concepts and programming techniques.
- Harvard College Research Program** Cambridge, MA
Researcher June 8-August 10, 2024
- Developed a transformer model to predict dietary index scores from metabolite data, identifying metabolic profiles linked to diet patterns and cardiovascular disease risk, potentially improving personalized nutrition and public health outcomes.
- Opportunity Insights** Cambridge, MA
Founding Research Assistant June-August 2024
- Contributed to the foundational research of the Spanish Opportunity Lab, a landmark initiative on intergenerational mobility in Spain.
 - Featured in El País, Spain's leading newspaper, with over 500,000 views and launched with remarks from former Prime Minister Felipe González.
- University of Minnesota School of Public Health** Minneapolis, MN
Researcher June 2017-August 2023
- Developed predictive models using advanced mathematical techniques to forecast individual outcomes based on longitudinal data, with applications in health, aging, and beyond. The models were trained on large datasets spanning multiple decades and evaluated using robust statistical methods.

- The models are applicable to process high-dimensional data, handle missing information, and predict future scenarios based on a variety of initial conditions.

Analytic Focus

Analytic Researcher

San Antonio, TX
May 30-June 27, 2023

- Envisioned and developed machine learning classification model in Python, resulting in a 70x boost in project efficiency.

QBE Insurance Group

People Analytics Intern

Remote
March 20-May 26, 2023

- Developed a comprehensive dashboard to track employee achievements and researched enhance workplace diversity.

Dangy Inc.

Full-stack developer

Remote
June 6-August 12, 2022

- Developed robust back-end solutions (Java) through personalized dish recommendations and streamlined app delivery services.

Junior Summer Math Camp at Texas State University

Counselor

San Marcos, TX
June 5-17, 2022

- Led group lessons, worked with campers to increase their problem-solving skills, responded to incidents, logged events, and discussed camper participation and concerns with all involved.

LEADERSHIP EXPERIENCE

Math Include (mathinclude.org)

Founder and CEO

International NPO
2018-Present

- Founded and lead one of the world's largest student-founded nonprofits driving youth development and educational equity globally.
- Delivered 48,312 free classes to 42,212 students across 1,054 schools in 44 U.S. states and 15 countries; founded 140 global chapters, mobilized 879 volunteers, created a 1,000-page curriculum, raised \$104K+, and awarded 1,641 internships and scholarships to historically underserved youth.
- Listed in Texas 211, expanding access to free STEM education and resources to 3 million underserved Texans.
- Partnered with the UN Major Group for Children and Youth and the Global Youth Constituency for Quality Education, bringing high-quality, inclusive learning to 300K+ students worldwide.

Charles River Economics Labs (Harvard, Stanford, U of Chicago, Brown)

Co-President

Cambridge, MA
August 2023-Present

- Lead a national premier student think tank conducting high-impact, pro bono economic research for governments, IGOs, and nonprofits to drive evidence-based policy.
- Mentor and train the next generation of economics researchers from top institutions.
- Directed statistical analyses for Malaysia's RM 1.2 trillion energy transition roadmap, shaping fiscal strategies for sustainable climate investment without increasing national debt.
- Spearheaded the launch of the Spanish Opportunity Lab, generating major media impact: coverage of our work on intergenerational mobility surpassed 500,000 views on El País, Spain's leading newspaper, and the event featured closing remarks from former Spanish Prime Minister Felipe González.
- Led regression analyses demonstrating the effectiveness of Project Reconnect, supporting the Washington D.C. Department of Human Services in reducing shelter incidents, positively impacting 4,000 homeless individuals and informing aid to over 700,000 residents.
- Collaborated with leading agencies including the U.S. Departments of Commerce and State, European Central Bank, World Bank, National Bank of Moldova, World Health Organization, UN Development

Programme, Economic Research Institute for ASEAN and East Asia, HUD, and Malaysia's Khazanah Research Institute to advance global equity, sustainability, and economic resilience.

WECode Conference (Women Engineers Code)

Cambridge, MA

Director of Programming

August 2023-Present

- Lead programming for the world's largest student-run women in tech conference, driving global engagement with 3,000+ attendees from 302 colleges across 24 countries.
- Curated a distinguished lineup of 50+ top women leaders in technology, including CEOs, founders, and engineering executives, to deliver keynotes and panels.
- Designed and executed a dynamic, inclusive agenda to inspire and equip the next generation of women in tech with tools for leadership, innovation, and impact.
- Strengthened global community-building and representation in STEM by creating transformative learning and networking opportunities for women technologists worldwide.

National Center for Women and Information Technology

Cambridge, MA

Inaugural Harvard Campus Rep

August 2023-Present

- Appointed as Harvard's first NCWIT Campus Rep, representing 22,947 students and advancing gender equity in computing through national collaboration.
- Founded the first Boston-area NCWIT reception in partnership with Microsoft, convening 300+ attendees from 9 colleges and 11 high schools to foster intergenerational mentorship and networking.
- Serve as a key liaison to NCWIT's network of 575+ partner organizations, including corporations, universities, startups, and nonprofits, to drive inclusive programming and expand pathways for women in tech.
- Strengthen Harvard's presence within NCWIT's national community while catalyzing local efforts to support aspiring women technologists in the Greater Boston area.

National Celebration of Inclusive Excellence on Pi Day

Hybrid

General Chair

2019-Present

- Founded and lead a nationally recognized annual conference advancing diversity, equity, and inclusion in STEM, well known as the "Pi Day Conference."
- Curate keynote talks and panels featuring prominent mathematicians and scientists from historically excluded backgrounds, showcasing diverse pathways to success in STEM.
- Empowered and inspired 3,000+ K-12 students and educators across 40+ states through interactive programming, community dialogue, and equitable access to STEM role models.
- Created a replicable model for inclusive science celebration that centers student voice, identity, and belonging in academic achievement.

Harvard Undergraduate Data Analytics Group

Cambridge, MA

Associate

January 2024-Present

- Leverage statistical modeling tools in Python to deliver a data strategy and governance plan for the Coaching Association of Canada (CAC), managing privacy and scalability for 2 million coach records.
- Developed and implemented a long-term strategy for sustaining diversity data collection as the National Coaching Certification Program (NCCP) expands, ensuring comprehensive tracking of newly joined coaches and existing coach engagements.
- Identified and established key I&D metrics for monitoring progress within the coaching community, enhancing the Coaching Association of Canada's ability to measure and promote diversity and inclusion effectively.
- Formulated high-level data governance guidelines to advise the CAC on the ethical collection and monitoring of sensitive information, addressing privacy concerns while maintaining data integrity and confidentiality.

MIT Sloan Sustainability Initiative

En-Roads Climate Global Ambassador

Cambridge, MA

August 2023-Present

- Lead global advocacy using the En-ROADS climate solutions simulator to drive evidence-based climate action in legislative chambers, corporate boardrooms, community coalitions, and classrooms.
- Empower policymakers, business leaders, and students to make data-driven decisions by facilitating interactive climate workshops that explore the real-time impact of policy choices on global temperature outcomes.
- Amplify climate literacy and systems thinking across sectors, translating complex climate data into accessible, actionable strategies for achieving net-zero goals.
- Contributed to the global En-ROADS ambassador network's mission to engage over 180,000 stakeholders in over 130 countries, accelerating collective climate impact.

AI Closing the Digital Divide Move

Founder

San Antonio, TX

2020-Present

- Founded a citywide initiative to democratize access to artificial intelligence education for underserved students in San Antonio, the nation's largest Hispanic-majority city.
- Designed and launched the Beyond AI curriculum, introducing core AI concepts, ethics, and applications to 133 high schools and empowering over 103,140 students, many of them first-generation learners or from historically underrepresented groups.
- Led professional development workshops for teachers to integrate AI into their classrooms, enhancing long-term impact and ensuring equitable tech fluency across the public education system.
- Pioneered one of the first scalable, inclusive AI education models aimed at closing racial, socioeconomic, and geographic gaps in tech literacy and future workforce access.

Texas Student Bill of Demands

Legislative Coordinator

San Antonio, TX

2020-2023

- Spearheaded policy research and youth-led advocacy efforts to address systemic educational and economic barriers facing underserved students in Bexar County.
- Successfully championed the consolidation of \$12.5 million in funding to expand education-to-employment pipelines and reduce the number of 33,613 "opportunity youth"—young people aged 16–24 who are not in school or employed.
- Helped shape a student-driven legislative agenda that informed local policymakers, mobilized community stakeholders, and elevated youth voices in education reform conversations statewide.

San Antonio Youth Commission

Chair

San Antonio, TX

2022-2023

Vice Chair

2020-2022

- Designed and led the inaugural SA Speak Up Teen Mental Health Survey, reaching thousands of youths across San Antonio in four languages to identify unmet mental health needs.
- Presented findings to city leadership, directly influencing the allocation of \$15 million in American Rescue Plan Act (ARPA) funds toward school-based mental health microgrants, peer training programs, and annual youth summits.
- Championed equitable access to mental health resources, creating a sustainable citywide model for youth-informed policy and cross-sector collaboration.

San Antonio Youth Success Consolidated Funding

Panelist

San Antonio, TX

2021-2023

- Appointed to represent 176,635 youth in Bexar County, ensuring youth voices were centered in public investment decisions.

- Evaluated and recommended funding allocations for high-impact youth success programs focused on education, workforce readiness, and community well-being.

Mu Alpha Theta | CS Club | Math Club | Science Fair Club

San Antonio, TX

Founder and Presidents

2019-2023

- Launched the Annual BASIS Middle School Math Competition, engaging hundreds of students in rigorous problem-solving and community building.
- Established the national award-winning BASIS Math Team and international championship-winning BASIS Science Fair Team, elevating the school's academic profile and inspiring a culture of achievement in STEM.

Editor-In-Chief & Blogger

2020-2023

- Founded and led Mathemagic, the BASIS Math Magazine, creating a platform for students to explore mathematical ideas and celebrate diverse voices in STEM.
- Edited and published student contributions, and authored invited articles on equitable math education featured on national education blogs, amplifying advocacy for inclusive and accessible STEM learning.

Basis National Honor Society

San Antonio, TX

Project Coordinator

2019-2020

- Coordinated service initiatives across 21 BASIS campuses nationwide, supporting 63 NHS and 21 NJHS members in delivering 9,976 collective volunteer hours.
- Scaled a unified service model that empowered students to contribute meaningfully to their communities while fostering leadership, collaboration, and civic responsibility on a national level.

SCHOLARSHIPS

2025 CRA Undergraduate Research Scholarship

\$1500

Recognized as one of the top 8 undergraduate researchers worldwide by the Computing Research Association.

2025 50 New Faces of Tech Scholarship

\$1500

Recognized as one of 50 emerging global female leaders in tech by 1000 Dreams Fund and HARMAN International.

2025 D. E. Shaw Discovery Fellowship

\$1500

Selected as a group of extraordinarily talented students to participate in an exclusive program in New York City.

2024 First-Place Winner in the ACM Student Research Competition

\$500

Winner in the ACM Student Research Competition (SRC) in the 2024 ACM Tapia Conference and selected as one of 15 global finalists for the 2025 ACM SRC Grand Finals.

2024 NCWIT Collegiate Award Winner

\$10,000

National Top 1 female college researcher in Computing for demonstrating exceptional creativity and impact through innovative technical projects, selected by the National Center for Women & Information Technology.

2024 Harvard College Research Award Winner

\$5,000

Awarded for exceptional interdisciplinary research contributions in computing, statistics, health sciences, and policy.

2024 Harvard First-Year Veritas Award Finalist	Top 3 Harvard Freshmen
Selected as one of three finalists out of 1,942 freshmen, recognized for demonstrating exceptional intellectual curiosity and commitment to academic excellence at Harvard College.	
2024 Third-Place Winner in Citadel Women's Datathon	\$2,500
Collaborated with a team of four to apply machine learning and time-series forecasting to analyze how gas price fluctuations correlate with macroeconomic factors and sub-industry performance.	
2023 Coolidge Presidential Scholar	Full Ride
A Full Ride scholarship, by Calvin Coolidge Presidential Foundation, applies to any university with any major. Selected for demonstrating an uncommon academic depth, intellectual curiosity, and exhibiting a genuine commitment to humanity and service. 5 scholars, 12,000 entrants.	
2023 U.S. Presidential Scholar	Top National Honor
Nation's top honor for high school students. Selected for passion and intellect, pursuit of excellence, and demonstrated commitment to high ideals. 161 scholars, 3.7M entrants.	
2023 Cameron Impact Scholar	Full Tuition
A Full Tuition Impact-Drive Scholarship applies to any university with any major. Selected for demonstrating a true passion for making a tangible and positive impact on my community and showing strength in academics. 15 scholars, 3000 entrants. Declined in favor of the Coolidge Presidential Scholarship.	
2023 82nd Regeneron Science Talent Search Top 40 Finalist	\$27,000
Nation's oldest and most prestigious pre-collegiate science competition. The first finalist in 33 years in San Antonio, TX, the 7th largest city in the US. 40 finalists, 1960 entrants.	
2023 National Top 5 Young Women in Mathematics Winner	\$1,200
National Top 1 girl and Top 4 overall in the American Mathematics Competition. 3X USA Mathematical Olympiad qualifier. Recipient of the AMC Jane Street Award, AMC Maryam Mirzakhani Award, and AMC Akamai Award.	
2023 Coca-Cola Scholar	\$20,000
Nation's largest corporate-sponsored, achievement-based scholarship program. 150 scholars, 91,000 entrants.	
2023 ACM Cutler-Bell Prize in High School Computing Winner	\$10,000
World's largest computing award for high school students. 1 of 4 global winners.	
2023 IvyMind Above and Beyond Scholar	\$25,000
Recognized for exceptional academic achievement, exemplary leadership, and a deep commitment to intellectual growth beyond the standard curriculum.	
2022 Cameron Impact Scholar	Full Tuition
A full tuition scholarship applies to any university and any major. Selected for exceptional leadership, service, and demonstrated commitment to creating lasting social impact. Declined.	
2022 Atlas Fellow	\$50,000

A global talent search for original thinkers of all backgrounds. Selected for dedication to solving the world's most pressing problems in math and computer science.

2021 Gloria Barron Prize for Young Heroes Winner **\$10,000**

Awarded for making a significant positive difference to people and the environment. 15 winners, 3000 entrants.

2021 Spirit of Ramanujan Fellow **\$5,000**

A global STEM talent search recognized alongside the MacArthur Fellowship.

2021 World Science Scholar **Two-Year Program**

Two-year learning and discovery program alongside Nobel and Breakthrough Prize winners. 48 scholars from 25 countries.

2021 Regeneron International Science and Engineering Fair (ISEF) Winner **\$1500**

ISEF 4th Place Grand Prize Winner (World Top 20) and 2nd Place American Statistical Association Special Award Winner.

2021 Princeton Prize in Race Relations Winner **\$1,000**

Awarded for significant efforts to advance racial equity, stand up to intolerance, and promote racial harmony in my school and community.

2021-2022 Carson Scholar **\$1,000**

Awarded for excelling academically and dedicating myself to serving my community.

2021-2022 Trinity University Scholarship **College Credits**

Covered all 21 credits taken at Trinity University as a high school student.

ACTIVITIES

Keynote Speaker at Harvard Youth Public Health Symposia **April 20, 2025**

Invited as the sole undergraduate from Harvard College **Cambridge, MA**
to inspire students to explore the intersection of science, computing, and community service.

Citadel Security Trading Invitational **April 3-4, 2025**

Selected from a highly competitive applicant pool to apply quantitative **New York City**
strategies and market-making techniques in a real-world trading simulation.

SIG Sophomore Discovery Day - Trading **March 27-28, 2025**

Selected for SIG Discovery Day, a competitive program at **Philadelphia PA**
Susquehanna International Group (SIG), gaining hands-on experience in options trading,
market-making, and game theory applications in quantitative finance.

D. E. Shaw Discovery Fellowship **January 6-8, 2025**

Selected as a group of extraordinarily talented students. Learned innovation in **New York City**
quantitative finance, computational problem-solving, and cutting-edge software development while
engaging with industry leaders and a cohort of exceptional peers.

Susquehanna Women's Discovery Day **November 15, 2024**

Selected from a competitive pool of applicants to attend Susquehanna Women's Discovery Day, **Remote**

an exclusive event fostering leadership and career development for women in finance and technology.

Heidelberg Laureate Forum Young Researcher

September 21-28, 2024

Selected as a World Top 200 Young Researcher to engage with Nobel Laureates and winners of the Abel Prize, Turing Award, and Fields Medal. Received a full fellowship to attend the prestigious forum in Heidelberg, Germany.

ACM Tapia Scholar

September 18-20, 2024

Awarded a fully funded scholarship to present a paper in the 2024 ACM Tapia conference. San Diego, CA

D. E. Shaw Connect

September 10, 2024

Selected to participate in the exclusive D. E. Shaw Connect event, recognized for outstanding potential in quantitative finance and leadership. New York City

Coolidge Summer Seminar

June 28-July 4, 2022-2027

Met the Trustees of Calvin Coolidge Presidential Foundation and received training in leadership, debate, public speaking, and politics. Plymouth, VT

Discovery Citadel and Citadel Securities

April 18-20, 2024

Selected for the prestigious Discover Citadel and Citadel Securities program, demonstrating exceptional analytical skills and financial acumen in a competitive applicant pool. New York City

Jane Street FTTP (First-Year Trading and Technology Program)

March 25-29, 2024

Invited as a top-performing first-year student skilled in advanced quantitative analysis and innovative trading strategies within a high-stakes financial setting. New York City

AAAI Undergraduate Consortium

February 20-27, 2024

Selected as a top AI undergraduate researcher by the Association for the Advancement of Artificial Intelligence. Awarded a scholarship to present paper in the 38th AAAI Conference, the top conference in AI. Vancouver, Canada

3rd-Place Winner in Citadel Women's Datathon

February 9-11, 2024

Collaborated with a team of four to apply machine learning and time-series forecasting to analyze how gas price fluctuations' correlation with macroeconomic factors and sub-industry performance. Miami, FL

SIG Discovery Day

February 8, 2024

Invited as a top-performing first-year student. Gained insights into the world of proprietary trading, explored quantitative strategies, and networked with SIG professionals. Remote

Two Sigma New Seekers Summit

January 18-19, 2024

Invited as a high-achieving student interested in exploring innovative solutions in technology and finance. Engaged in collaborative workshops, networked with industry leaders, and gained insights into cutting-edge data science and algorithmic trading strategies. Remote

2nd-Place Winner in Harvard First-Year Entrepreneurship Pitch Competition

August 4, 2023

Pitched BrainBank, a marketplace that delivers specialized service and networks reliably and affordably. Cambridge, MA

Harvard First-Year Urban Pre-Orientation Program August 24-31, 2023
 Admitted with a 27% acceptance rate, enhancing community awareness Cambridge, MA
 through urban exploration and service projects.

Mathematical Olympiad Awards Ceremony August 4, 2021 & August 6, 2023
 Two times recognized for a National Top 5 Young Women in Mathematics Winner. Miami, FL
 Featured on the Mathematical Association of America main page (maa.org) and Math Values.

Canada/USA Mathcamp July 5-August 9, 2020 & July 3-August 7, 2022
 Researched advanced topics in math modeling and machine learning. Waterville, ME

Stanford University Mathematics Camp July 10- July 30, 2021
 Researched number theory and coding theory. Stanford, CA

Summer@SaMi Camp 2019-Present
 Organized the nation's largest online K-12 STEM camps for 13,000+ students. Featured in ABC, CBS &
 Spectrum News.

Atlas Fellow Summer Camp August 14-25, 2022
 Gained tools to solve global issues and developed a framework for change. Berkeley, CA

Coolidge Senator Summit August 25-28, 2022
 Heard from policy leaders and visited the White House. Washington DC

National Points of Light Conference July 14-16, 2021
 Invited to attend the 2021 National Points of Light Conference and learned innovative ideas Remote
 regarding future global nonprofits and education equity.

President Ronald Reagan Student Leadership Program August 2-6, 2021
 Accepted to the SLP Program and learned skills towards improving our lives Remote
 through public service and policymaking.

City's Advocate for Youth Vaccination May 28-June 2, 2021
 Led the advocacy for vaccination for youth aged 16-24 in San Antonio. San Antonio, TX

AMERICAN MATH/PHYSICS/COMPUTING COMPETITION SCORES

- 3X USA Math Olympiad (USAMO): 4/42 (2023), not taken (2022), 1/42 (2021).
- 2X USA Physics Olympiad (USAPHO): AAPT doesn't release scores (2023, 2022).
- USA Computing Olympiad (USACO) Gold: 574/800 (2020).
- 4X American Invitational Math Exam (AIME): 11/15 (2023), 11/15 (2022), 10/15 (2021), 8/15 (2020).
- American Mathematics Competition (AMC) 10B: 144/150 (2021 National Top 1 girl and Top 4 overall).
- AMC 12A: 145.5/150 (2021 National Top 5 girls and AMC Jane Street Award).
- AMC 10A: 135/150 (2020 AMC Maryam Mirzakhani Award).
- AMC 12B: 144/150 (2023 National Top 5 girls and AMC Akamai Award).
- F=MA: 22/25 (2023 National Top 0.3%), 18/25 (2022 National Top 1%).
- USA Math Talent Search Bronze Prize: 48/75 (2021).
- 3X MIT Math Prize for Girls (2022, 2021, 2020).
- US Team and Co-Champions of 2018 Primary Math World Competition: 15/15 (World Top 4).

RESEARCH TALKS

- **Guan H**, Advancing Subseasonal Forecasting with Machine Learning, AI Weather Quest Year 1 Awards Ceremony, September 24, 2026.
- **Guan H**, Frontiers of Trustworthy AI, Trinity University Computer Science Colloquium, San Antonio, TX, March 19, 2026.
- **Guan H** and Jonathan Weyn, Probabilistic Adaptive Bias Correction for Sub-Seasonal Weather Forecasting, *AI Weather Quest Awards Invited Talk*, Virtual, December 18, 2025.
- **Guan H**, Exploring AI Methods for Developing Metabolomic Signatures of Nutritional Factors, *American Society of Human Genetics Annual Meeting*, Boston, MA, October 15, 2025.
- **Guan H**, Interpretable Deep Learning for Aging Dynamics and Mortality Prediction, American Society of Human Genetics Annual Meeting, Boston, MA, October 16, 2025.
- **Guan H**, Interpretable High-Dimensional Networks for Aging and Survival Modeling, *CRA Undergraduate Research Awardee Lightning Talks with National Lab Sponsors*, Virtual, May 12, 2025.
- **Guan H**, HiDiNet: High-Dimensional Interpretive Network for Modeling Aging Health and Survival, *ACM Tapia Conference*, San Diego, CA, September 19, 2024.
- **Guan H**, Innovative Method for Longitudinal Analysis of Aging Health and Mortality Rate Prediction, *Joint Statistical Meetings*, Portland, OR, August 6, 2024.
- **Guan H**, A Novel Approach for Longitudinal Modeling of Aging Health and Predicting Mortality Rates, *AAAI Conference on Artificial Intelligence Undergraduate Consortium*, Vancouver, Canada, February 24, 2024.
- **Guan H**, Biological Aging Prediction in Neural Networks, *Joint Statistical Meetings*, Washington DC, August 8, 2022.
- **Guan H**, The Genetics of Human Aging: Predicting Age and Age-Related Diseases by Deep Mining High Dimensional Biomarker Data, *Sigma Xi Student Research Showcase*, Virtual, April 25, 2022.
- **Guan H**, Interpreting Epigenetic Aging with Deep Neural Networks Using High Dimensional DNA Methylation Data, *Joint Mathematics Meetings*, Virtual, January 7, 2022.
- **Guan H**, The Genetics of Human Aging: Predicting Age and Age-Related Diseases by Deep Mining High Dimensional Biomarker Data, *Sigma Xi Annual Meeting and Student Research Conference*, Virtual, November 4, 2021.
- **Guan H**, Interpreting Epigenetic Aging with Deep Neural Networks Using High Dimensional DNA Methylation Data, *American Society of Human Genetics Annual Meeting*, Virtual, October 18, 2021.
- **Guan H**, The Genetics of Human Aging: Predicting Age and Age-Related Diseases by Deep Mining High Dimensional Biomarker Data, *Mathematical Association of America MathFest*, Virtual, August 5, 2021.
- **Guan H**, Genetic Prediction of Biological Age: Exploring the Relationship between Epigenetic Markers and All-Cause Mortality, *Joint Mathematics Meetings*, January 8, Virtual, 2021.
- **Guan H**, Predicting Biological Age: Neural Network and Algorithmic Models for High Dimensional DNA Methylation Data, *American Society of Human Genetics Annual Meeting*, October 27, Virtual, 2020.

PROFESSIONAL REVIEW

- NPJ Aging.
- ICLR 2026 Workshop Re-Align.
- COLM 2026.

MEDIA COVERAGE

- **Daily Papers and Machine Learning Papers** both featured our Paper "Enhancing AI and Dynamical Subseasonal Forecasts with Probabilistic Bias Correction" on April 20, 2026.
<https://x.com/Memoirs/status/2046203438720078268>
- **AAAS EurekAlert, The Results Are In! ECMWF's AI Weather Quest Concludes Latest Period**
<https://www.eurekalert.org/news-releases/1119902>
- **ECMWF, The AI Weather Quest: Spotlighting The Best of Machine Learning Sub-Seasonal Prediction** <https://www.ecmwf.int/en/about/media-centre/science-blog/2026/ai-weather-quest-2026>
- **MIT News, Decoding the Arctic to Predict Winter Weather** <https://news.mit.edu/2026/decoding-arctic-to-predict-winter-weather-0108>
- **Foundational Research of the Spanish Opportunity Lab Featured in El País, Spain's Leading Newspaper with 500,000 views** <https://elpais.com/economia/2025-05-14/mapa-de-la-renta-de-padres-e-hijos-en-que-ciudades-funciona-mejor-el-ascensor-social.html>
- **Harvard School of Engineering and Applied Sciences – The Community beyond the Conference**
<https://seas.harvard.edu/news/2025/03/community-beyond-conference>
- **Testimony for the ACM Student Research Competitions** <https://src.acm.org/testimonials/2025>
- **Heroes Among Us – Carson Foundation Original Movie**
<https://www.youtube.com/watch?v=rrMP9DSWKvo>
- **Amazing Americans – Interviewed by Congressman Will Hurd**
<https://www.facebook.com/HurdOnTheHill/videos/2514790241970714/?v=2514790241970714>
- **Authority Magazine – Young Change Makers: Why and How Hannah Guan Is Helping to Change Our World** <https://medium.com/authority-magazine/young-change-makers-why-and-how-hannah-guan-is-helping-to-change-our-world-d18cd878fef6>
- **CBS – Girls Role Model in Math** <https://youtu.be/trLTrj62EFM>
- **CBS – A 14-yr-old Fights for COVID-19 Summer Slide** <https://youtu.be/dX53oy-4tp0>
- **CBS – Kids Who Make San Antonio Great** <https://www.kens5.com/article/news/local/outreach/kids-who-make-sa-great/kids-who-make-sa-great-13-year-old-non-profit-ceo-on-a-mission-to-make-math-students-better/273-5e10470e-c79f-4d18-8eda-e5860819d2d7>
- **ABC – San Antonio High School Student Wins National Award for Helping Bring STEM Education to Underserved Communities** <https://www.ksat.com/news/local/2022/10/05/san-antonio-high-school-student-wins-national-award-for-helping-bring-stem-education-to-underserved-communities/>
- **ABC – San Antonio High School Senior Named Finalist in Renowned Science Competition**
<https://www.ksat.com/news/local/2023/01/24/san-antonio-high-school-senior-named-finalist-in-renowned-science-competition/>
- **Axios – San Antonio Teen Competing in Prestigious Science Event** <https://www.axios.com/local/san-antonio/2023/02/03/hannah-guan-san-antonio-regeneron-science>
- **Spectrum News – Texan Wins National Award for Creating Free Math Lessons**
<https://fb.watch/gBFYmp9dNS/>
- **BASIS.EDU Outcomes Magazine – Building the Business of STEM Education Worldwide**
https://drive.google.com/file/d/17PuvuzwlucqXZZep-DV_kOugQC3Thxa5/view?usp=sharing
- **Mathematical Association of America**
<https://www.facebook.com/maanews/photos/a.170415820418/10159403413345419>
- **Mathematical Association of America Math Values** <https://www.mathvalues.org/masterblog/the-spark-ignited-by-amc-finding-and-spreading-the-beauty-in-math>
- **American Statistical Association News Magazine, December 2021 Issue**
https://magazine.amstat.org/wp-content/uploads/2021/12/December-2021_2.pdf
- **Association of Computing Machinery Digital Library Featured Research**
<https://twitter.com/ACMDL/status/1427632086811299842>

- **National Center for Women and Information Technology – Celebrating AAPI Women in STEM:**
Hannah Guan, Aging Research Innovator <https://x.com/NCWIT/status/1796241695857618976>
- **Julian C. Stanley Study of Exceptional Talent Newsletter**
<https://drive.google.com/file/d/1sH8A9kKwBV5bV8EOuxmIuoE-6aENUU7m/view?usp=sharing>
- **Alamo Regional Science and Engineering Fair Keynote Speaker**
<https://www.youtube.com/watch?v=FO52YVGf9VU>
- **Mathworks Alumni Feature Stories** <https://www.txstate.edu/mathworks/about/Alumni-Stories.html>
- **Texas Education Agency** <https://twitter.com/teainfo/status/1360272568767414278>
- **TEDx@San Antonio** <https://tedxsanantonio.com/your-ideas-matter/>
- **Rivard Report** <https://sanantonioreport.org/learning-loss-during-covid-19-shutdown-adds-to-summer-slide/>
- **City of San Antonio** <https://www.youtube.com/watch?v=whJI2m6wLQw>
- **San Antonio's Extraordinary Women** <https://www.msn.com/en-us/news/us/san-antonios-extraordinary-women-hannah-guan/ar-AAVA9TD>
- **San Antonio Express-News** <https://www.expressnews.com/news/local/article/15-year-old-San-Antonio-mathematician-OLLU-grad-16536064.php#photo-21587268>
- **San Antonio Current** <https://www.sacurrent.com/news/san-antonio-high-schooler-named-finalist-in-prestigious-national-science-competition-30897320?fbclid=IwAR26SZKl4Njlejw1A1Ppfu36cqRaC5UFc7eURzYWOSwYdiYra0TUOa3syDQ>
- **San Antonio Current** <https://www.sacurrent.com/news/san-antonio-high-schooler-wins-national-award-for-helping-students-from-underserved-communities-29922847>
- **San Antonio Charter Moms** <https://sachartermoms.com/learn-love-math-change-world/>